Second Technical Response to the Revised Manuscript

Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction

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Abstract

This note responds to the revised April 2026 version of the two-tier quotient paper. The revision makes real progress, and I withdraw several objections from my earlier response. In particular, the switch from $Core(Desc)$ to $Union(Desc)$ for non-PP designated witnesses, the addition of the type-safety filter $Safe$ to $V6$, the indexing of kernels by $(descriptor, external\ interface)$ pairs, and the clarification that kernel PP-loops are certificate devices rather than model diagonals are all substantial improvements. I therefore no longer regard those earlier points as decisive objections. I still do not think, however, that the main decidability theorem is proved. Four issues remain: (i) phase-specific off-period $\exists PP$ demands are still not covered; (ii) constant kernel-to-external relations are too coarse to preserve phase-specific universal restrictions; (iii) the blocking claim used in the size bound is not established for regular nodes; and (iv) the passage from the finite $V6$ certificate to the unfolded infinite model is not proved. My present judgment is therefore: meaningful progress, but not yet a completed proof of decidability.

1. Overall judgment

The revised manuscript is materially stronger than the original version. The revision note correctly identifies several serious objections and addresses them in technically sensible ways. I therefore want to record at the outset that this is not a repetition of my previous review. Some earlier criticisms should now be treated as answered.

At the same time, I do not think Theorem 9.1 is established yet. The remaining issues are not cosmetic. They occur exactly where the manuscript must succeed in order to convert a finite quotient into a genuine $\mathcal{ALCI}_{RCC5}$ model. Since Wessel's report left the decidability of $\mathcal{ALCI}_{RCC5}$ open, any affirmative claim has to close those points explicitly and not only heuristically [3].

2. What the revision fixes

The revised paper addresses several of the strongest objections to the original draft. The current status, in my view, is summarized in Table 1.

Earlier objection	Status	Comment
Non-PP designated witnesses were required only for $Core(Desc)$	Repaired	Definition 6.2 now uses $Union(Desc)$ for non-PP demands, which fixes the earlier phase-loss objection on that part of the construction.
$V6$ checked only RCC5 composition and ignored type safety	Repaired	Definition 2.1 introduces $Safe(\tau_1, \tau_2)$ and $V6$ now intersects composition domains with that filter.
One kernel per descriptor forgot stabilized external behavior	Largely repaired	Kernels are now indexed by $(descriptor, external\ interface)$ pairs, which is the right direction and addresses the original descriptor-only objection.
Kernel PP-loops were treated as if they were model diagonals	Clarified	Remark 6.3 now presents the quotient as a certificate rather than as an ordinary strong-EQ RCC5 model.

Table 1: Points on which the revised manuscript improves substantially over the original.

Because of these changes, my present response is narrower than my earlier one. The remaining criticisms are the ones that, in my view, still block Theorem 9.1.

3. Remaining technical problems

3.1 Phase-specific off-period PP demands are still missing

The switch from $Core(Desc)$ to $Union(Desc)$ was made only for *non-PP* designated witnesses in $T4$. By contrast, the off-chain extension condition $V6$ still quantifies over

"each $\exists PP.C$ demand of a kernel node k_α that is not satisfied within $Desc_\alpha$."

But the kernel type is still

$$tp(k_\alpha) = Core(Desc_\alpha),$$

so a phase-specific $\exists PP.C$ that is present in some phase type τ_i but not in the core is invisible at the level of the kernel.

This matters because Theorem 8.1, Step 5, later claims that for *every* phase type τ_i and every $\exists PP.C \in \tau_i$, the demand is satisfied either within the period or by an off-chain PP witness from

$V6$. That conclusion does not follow from the present definition of $V6$.

A schematic counterexample is immediate. Let a descriptor have two phase types τ_A, τ_B such that

$$\exists PP.B \in \tau_A, \qquad \exists PP.B \notin \tau_B,$$

and suppose no phase in the descriptor contains B. Then

$$\exists PP.B \in Union(Desc) \setminus Core(Desc).$$

The current $V6$ never asks for an off-chain PP witness for that demand, because the kernel node does not carry it. Yet Step 5 of Theorem 8.1 later relies on exactly such a witness.

So the revision repaired the non-PP side of the earlier objection, but it did not repair the analogous problem for phase-specific off-period PP demands.

3.2 Constant kernel interfaces are too coarse for phase-specific universals

Theorem 8.1, Step 2 assigns every sufficiently far chain element the same external relation as its kernel:

$$\rho(d_i^{(\alpha)}, e) = \rho(k_\alpha, e).$$

Thus an external designated witness or regular node is related in the same RCC5 way to *every* phase on that periodic tail.

However, $T4$ checks type safety only for those phase types that actually contain the existential being witnessed. Concretely, if w is a designated witness for a demand $\exists R.C$, the revised $T4$ requires

$$R \in Safe(\tau_j, tp(w))$$

for every phase type τ_j that contains that demand. It does *not* require safety against other phases of the same descriptor that do not contain the existential but do carry universal restrictions about R.

This is still too weak. Consider a descriptor with phase types τ_A, τ_B satisfying

$$\exists DR.A \in \tau_A, \qquad \forall DR.\neg A \in \tau_B.$$

Let w be a designated DR-witness with $A \in tp(w)$. The revised $T4$ can approve this witness, because the safety check is performed only against τ_A, the phase that contains the existential demand. But in the realization, Step 2 makes every τ_B-position on the tail also DR-related to w. That violates $\forall DR.\neg A \in \tau_B$.

So the current quotient still forgets too much phase-sensitive external information. The likely repair is to index external interfaces by phase, or else to strengthen the validity conditions so that every inherited external relation is checked against every phase to which it will be copied.

3.3 Claim 7.2 does not justify the blocking bound for regular nodes

The revised manuscript blocks off-chain witnesses by a key consisting of

1. the witness type,
2. the demanded relation to the parent, and
3. the stabilized relations to all kernel nodes.

Claim 7.2 then states that witnesses with the same key have identical $V6$ domains with respect to every existing node e.

That claim is not proved for regular nodes, and as stated it is generally false. The domain formula is

$$D(w, e) = comp(R_{dem}, \rho(parent, e)) \cap Safe(tp(w), tp(e)).$$

For a kernel node e, the term $\rho(parent, e)$ is controlled by the parent's kernel interface. But for a *regular* node e, two parents may agree on all kernel relations and still disagree on $\rho(parent, e)$.

This already breaks the claim before the $Safe$ filter is even considered. For example, let two parents p, p' have the same type and the same relations to all kernel nodes, but let some regular node e satisfy

$$\rho(p, e) = DR, \qquad \rho(p', e) = PO.$$

Let the demanded relation be $R_{dem} = PP$. Then the two domains differ because

$$comp(PP, DR) = \{DR\}, \qquad comp(PP, PO) = \{DR, PO, PP\}.$$

Hence two children can share the stated blocking key and still have different $V6$ domains relative to the same regular node e.

Since Claim 7.2 is the engine that supports subtree redirection and the computable size bound in Theorem 7.1, the current proof of the bound is incomplete. The manuscript still owes a blocking theorem strong enough to justify identification in the presence of regular-to-regular interaction.

3.4 The quotient-to-model lifting step is still missing

Theorem 8.1, Step 4 appeals to full RCC5 tractability after saying that the extension network, "including the unfolded chain elements," is satisfiable. But $V6$ only checks path-consistency for a *finite* network obtained by adding off-chain witness variables to the kernel/regular quotient.

What is missing is a lift lemma: a theorem showing that the finite $V6$ certificate remains path-consistent after kernels are replaced by infinitely many phase-labelled chain elements. At present, that does not follow formally from the definitions. In the unfolded model there are many more endpoints, the phases on a tail need not all have the same type, and Step 2 copies one kernel relation

uniformly across positions whose universal theories may differ.

So even if one grants the finite $V6$ check, the manuscript still needs to prove that this finite check is the right certificate for the infinite unfolded structure to which Theorem 2.3 is then applied [2]. Without that bridge, Step 4 remains an assertion rather than a theorem.

4. Consequence for Theorem 9.1

Because of the revisions, I would now withdraw several earlier objections. But the four points above remain exactly where the paper must succeed in order to prove decidability:

- Theorem 7.1 needs a correct finite bound on quotient size.
- Theorem 8.1 needs a realization argument that preserves every existential and universal obligation of every phase.
- Theorem 9.1 depends on both.

In my judgment, those obligations are not yet discharged. I therefore would not say that the revised manuscript closes Wessel's open problem. I would say instead that it narrows the gap substantially and sharpens the remaining proof obligations.

5. What would make the proof convincing

A further revision would become much more persuasive if it did the following explicitly:

1. extend the $Union(Desc)$ idea to the off-period PP case as well, so that every phase-level $\exists PP$ obligation is represented in the quotient;
2. replace constant kernel interfaces by phase-sensitive external interfaces, or prove an equivalent phase-uniformity theorem strong enough for Step 2;
3. strengthen the blocking key, or prove a new claim that really controls regular-node interaction and yields a non-circular size bound;
4. add a quotient-to-unfolding lift lemma showing that the finite $V6$ certificate induces a path-consistent disjunctive network on the infinite unfolded structure.

Those are substantial tasks, but they are now well localized. That is real progress.

6. Conclusion

My updated assessment is therefore as follows. The revised paper is a genuine improvement over the original version, and several of my earlier criticisms should now be considered answered.

Nevertheless, I still do not think the paper proves decidability of $\mathcal{ALCI}_{RCC5}$. The remaining problems are concentrated, serious, and mathematically precise. The manuscript is closer than before, but it has not yet crossed the line from a promising construction to a completed theorem.

References

[1] Claude. *Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction: Period descriptors, PP-kernel nodes, and the full tractability of RCC5.* Revised discussion draft, April 2026.

[2] Jochen Renz and Bernhard Nebel. On the complexity of qualitative spatial reasoning: A maximal tractable fragment of the Region Connection Calculus. *Artificial Intelligence*, 108(1–2):69–123, 1999.

[3] Michael Wessel. *Qualitative Spatial Reasoning with the ALCIRCC Family – First Results and Unanswered Questions.* Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.